

1 UNIVERSITY OF CHINESE ACADEMY OF SCIENCES

2 DOCTORAL THESIS

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4 **The study of Higgs properties in the**
5 **four-lepton final state at CMS**

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14 **Contents**

15 **List of Figures** **v**

16 **List of Tables** **vii**

17 **1 The CMS Detector and Event reconstruction** **1**

18 1.1 The Compact Muon Solenoid (CMS) detector 1

19 1.1.1 General aspects and the geometry 2

20 The CMS magnet 3

21 1.1.2 CMS Coordinate system 4

22 1.1.3 Inner Tracking System (ITS) 5

23 The Pixel Detector 5

24 The Silicon Strip 7

25 1.1.4 Electromagnetic Calorimeter (ECAL) 8

26 1.1.5 Hadron Calorimeter (HCAL) 11

27 1.1.6 CMS Muon System 13

28 List of Figures

29	1.1	The CMS detector schematic diagram	2
30	1.2	CMS magnet system.	4
31	1.3	CMS spherical coordinte system	4
32	1.4	Schematic diagram of CMS tracker system. Lines represent detector	
33		modules (before phase 1 pixel upgrades).	6
34	1.5	Schematic diagram of pixel detector's upgrade. Old pixel detector	
35		is shown by the lower half detector, and the new pixel detector is	
36		represented by the upper half.	6
37	1.6	Pixel detector's hit efficiency as function of instantaneous luminos-	
38		ity corresponding to 2016 data (left) and to 2017 data (right)	7
39	1.7	CMS ECAL pseudo-rapidity coverage.	8
40	1.8	CMS ECAL layout	9
41	1.9	CMS ECAL energy resolution curve with individual term contribu-	
42		tions.	10
43	1.10	CMS ECAL energy resolution.	10
44	1.11	CMS HCAL layout.	11
45	1.12	CMS muon system layout.	14

List of Tables

46

Chapter 1

The CMS Detector and Event reconstruction

Since the data used for the analysis in the dissertation was collected using the CMS detector, this chapter would focus on geometry of CMS, its parts (sub-detectors) and description of how events are reconstructed with the CMS detector.

1.1 The Compact Muon Solenoid (CMS) detector

LHC has set new frontiers for researches in particle physics experiments. State of the art experimental facility has annexed with equipment based on latest technology. To assure the validity of results from the experiment, LHC has its two major and general purpose experiments for cross-validation, one of which is CMS. They have same physics goals with some different design and the detector technology. Particularly the CMS is capable to provide good resolution for muon identification and momentum over wide range of dimensions and determination of the charge on muons with ultimate precision with momentum up to 1 TeV. It gives healthy inner tracker reconstruction efficiency and resolution of charged particle momentum. It has efficient trigger system and better discrimination of τ 's and b-jets. Further, it is benefitted from good dielectron, diphoton and dijet mass and energy resolutions, and as well as missing-transverse-energy (MET) resolutions. This makes it capable to study large areas from SM Physics (including Higgs Physics) to search for extra dimensions and the dark matter covering physics in energy range from a few 100 GeV to TeV scale [Reference34].

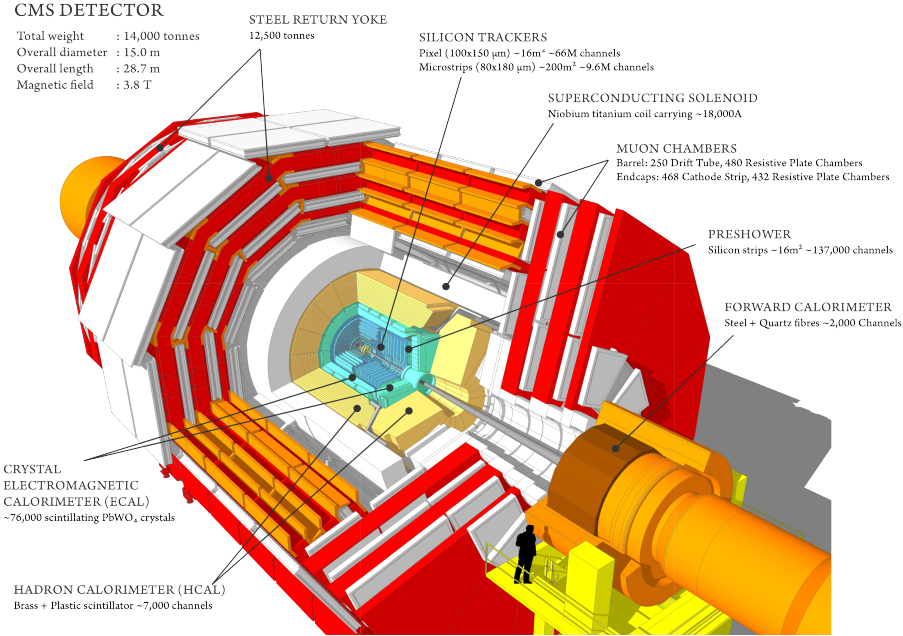


FIGURE 1.1: The CMS detector schematic diagram

1.1.1 General aspects and the geometry

Being one of the two general purpose detectors installed at LHC ring, CMS is constructed in cylindrical shape having length about 28 m and outer radius of around 7 m. Its overall height is about 15 m with overall weight of about 14,000 tonnes. Overall design of the CMS detector can be seen in the Figure 1.1. Learning from the earlier experiment at CERN, CMS engineered to be built in different slices at surface instead of doing everything underground. And fifteen separate slices were then lowered to underground to be fixed at the LHC tunnel.

Moving radially outwards from interaction point of beams, there are inner tracker system (ITS), electromagnetic calorimeter (ECAL) and hadronic calorimeter (HCAL) sub-detectors of CMS which are fitted inside the large solenoid magnet. Inner tracker of CMS has large track multiplicity as being closest to beam pipe where collisions happen. ECAL is built with crystals of lead tungstate which covers pseudorapidity up to $|\eta| < 3.0$. After ECAL, hadronic calorimeter is installed which is made of brass. Outermost layer of the CMS detector is the Muon System. Details of sub-detectors, trigger system and data acquisition systems will be described in next Sections.

The CMS magnet

Measuring the momentum of the charged particle with great precision is one of the ultimate goals of the CMS detector. In this regard, choice of magnetic field configuration is a very important feature of a particle detector like CMS. Stronger magnetic field is needed to measure high energy charged particle momentum with precision. In CMS, this is achieved by selecting the superconducting solenoid magnet.

When a charged particle moves in uniform magnetic field of a solenoid, its momentum resolution which depends upon strength of magnetic field and the radius of solenoid, can be written as:

$$\frac{dp}{p} \propto \frac{p}{Br^2} \quad (1.1)$$

Here r is the path length traversed by the charged particle inside the magnetic field (it becomes radius of the magnetic if the particle passed the full magnet). It is obvious from the relation, that for detector to obtain high momentum resolution, we need a volumetric solenoid magnet with stronger magnetic field. CMS solenoid magnet is designed to have bore of diameter 6 m and length 12.5 m producing magnetic field of 3.8 T which is 100,000 times stronger than earth magnet. It is the largest ever built magnet weighs about 12,000 tonnes. It is powerful enough in providing a strong magnetic field in its dimensions throughout the detector. It surrounds CMS ITS, ECAL and HCAL. The strong magnetic field efficiently bends the charged particles when they move through it. More the momentum of the charged particles less they bend, so high energetic charged particles require a stronger magnetic field for its momentum to be measured accurately. CMS aims to have excellent momentum resolution with such strong magnetic field specially for muons. The schematic diagram of CMS magnet is given in the Figure 1.2.

The CMS magnetic is made of coil of NbTi wire which becomes a source of uniform magnetic field when current passes through it. The NbTi becomes superconductor at -268.5°C (4 K) which allows a huge amount of current (10,000 A) to pass through it. To maintain this temperature liquid helium is used. The magnetic field direction inside (tracker) and outside the solenoid (muon system) allows to make second momentum measurement for muons [cms:MFTDR, Reference36, Reference37, Reference38].

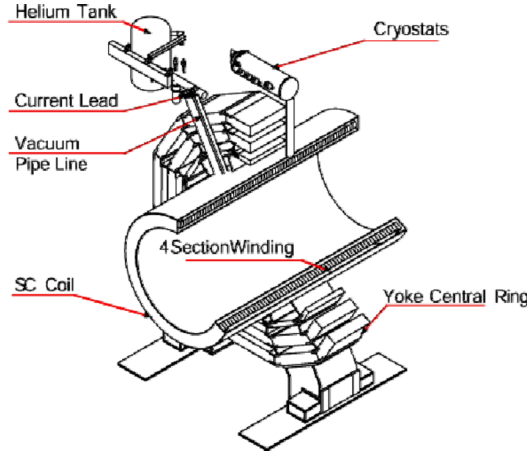


FIGURE 1.2: CMS magnet system.

1.1.2 CMS Coordinate system

Coordinate system used for CMS detector for its operations is sketched in Figures 1.3(a) ?? . The CMS detector is in cylindrical shape around beam axis as illustrated in Figure 1.1 which is represented by $Z - axis$ in Figures 1.3(a) 1.3(a). The point where two beams collides in beam pipe is called center of experiment; the y axis is directed vertically upwards and the x axis being horizontal towards the LHC ring center.

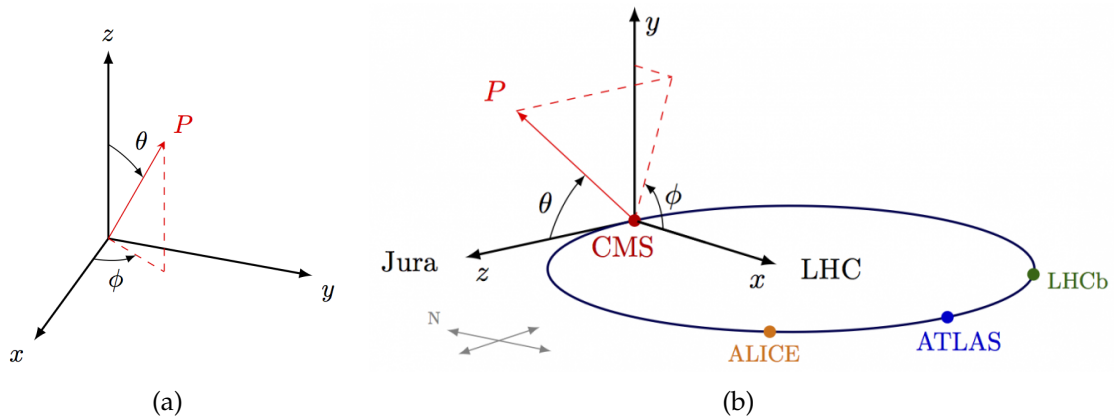


FIGURE 1.3: CMS spherical coordinte system

The r represents the radial component in transverse plane while ϕ is the the azimuthal angle measured from the x axis. θ which is measured from z axis is called polar angle. Position of particle when it passes through the CMS is described in terms of (η, ϕ) , where η being the pseudo-rapidity defined as $\eta = -\ln \tan(\theta/2)$ and

remains invariant under the longitudinal boosts in the environment of the partons which carry the fractions of the momentum from hard scattering collisions of e.g. proton-proton. Moreover, pseudorapidity also helps to differentiate the barrel regions from the endcap regions of sub detectors. In terms of pseudorapidity, the barrel region are defined by $|\eta| < 1.5$ whereas the endcap region corresponds to $1.5 < |\eta| < 3.0$.

1.1.3 Inner Tracking System (ITS)

The CMS ITS is a sub-detecting system of the CMS detector which is important from its task to give best tracks and hence the momenta (above 1 GeV) of the particles (electron, muon, hadrons-quark-structured particles and other short-lived particles) produced in the collisions that take place at CMS, LHC. It is the most inner sub-detector placed under influence of a strong and uniform magnetic field of CMS solenoid and it surrounds the region of beam pipe as shown in Figure 1.1. ITS also provides the position information of the primary and secondary vertices which is important to reject extra interactions and objects.

Being in cylindrical shape, it has a diameter of about 2.5 m and length about 5.5 m providing $|\eta|$ coverage up to 2.5. At LHC nominal luminosity of $10^{34} \text{ cm}^2 \text{ s}^{-1}$, there is an average of more than 20 proton-proton interactions (called pileup, for 2018 it was above 30) leading to average about 1000 particles emerging after every bunch crossing, 25 ns. ITS is entirely made up of (200 m^2 of) silicon which is benefitted from being super efficient in the environment of huge flux of particles coming out of the interaction point. A schematic diagram of CMS ITS is shown in the figure 1.4 (as stood before phase 1 pixel upgrade).

To limit the damage mainly by radiation ITS is kept at about -20° C . Full ITS is an assembly of a pixel detector and silicon strip tracker.

The Pixel Detector

Part of the tracker system which is nearest to the interaction point is the Pixel detector. Silicon modules of the pixel detector, are facilitated with pixel cells of size $100 \times 150 \mu\text{m}^2$ each providing the $|\eta|$ range up to 2.5. It helps to reconstruct the secondary vertex and delivers three spatial points precisely throughout its geometry [Reference34].

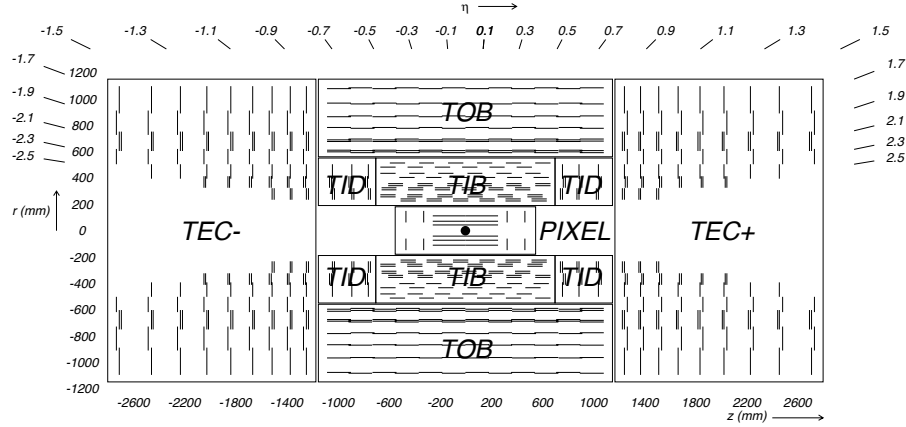


FIGURE 1.4: Schematic diagram of CMS tracker system. Lines represent detector modules (before phase 1 pixel upgrades).

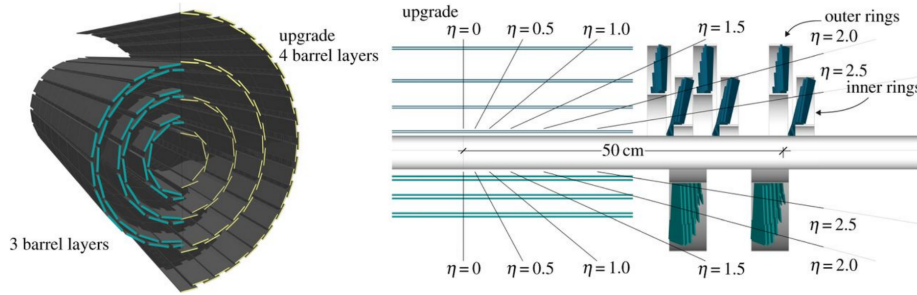


FIGURE 1.5: Schematic diagram of pixel detector's upgrade. Old pixel detector is shown by the lower half detector, and the new pixel detector is represented by the upper half.

During data taking of Run 1 operations of LHC, some dynamic inefficiencies (about 5% at an event pile-up of 40) were reported in the first layer of barrel of pixel sub-detector, due to the limited size of the readout bandwidth. Subsequently, a new pixel detector was installed in March 2017 Ref. ??, equipped with one additional barrel layer and one additional forward disk. In the new pixel detector, the innermost layer is closer to the interaction point and the outermost one is further away from it as it is shown in Fig. 1.5

It also uses new Readout Chips (ROCs) benefitted from reduced data loss at higher collision rates with respect to Phase 1 operations (at 160 Mbit/s). Bandwidth of the readout electronics was increased approximately by a factor of eight (320 Mbit/s). In addition, DC-DC power converters were installed to allow reusing the existing fibers and cables. Hence, the upgraded pixel detector has a higher rate capability and number of channels, providing increased tracking efficiency,

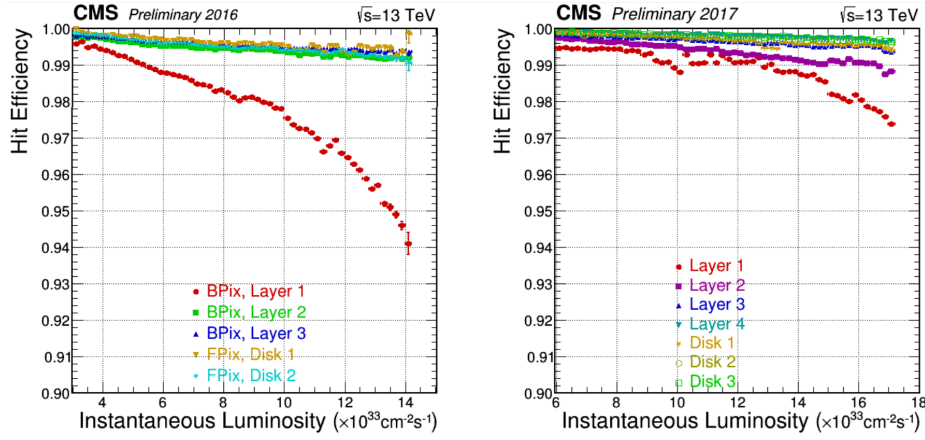


FIGURE 1.6: Pixel detector's hit efficiency as function of instantaneous luminosity corresponding to 2016 data (left) and to 2017 data (right)

reduced fake rate, and improved impact parameter resolution. In the figure 1.6, clear improvements in the dynamic efficiency of 2017 (with pixel upgrades) with respect to the 2016 (without pixel upgrades) can be seen Ref. [Sonneveld:2018ntf].

Overall, the pixel detectors is of 1 m^2 size containing 66 million pixels.

The Silicon Strip

Next layer to the pixel detector is called Silicon Strip Tracker which is installed at radial distance from 20cm to 116cm. This detector surrounds both sides of pixel with Its tacker inner barrel (TIB, 4 layers) and tracker inner disk (TID, 3 layers) surrounds the pixel detector and extend upto 20 cm-50 cm where silicon microstrips of thickness $320\mu\text{m}$ are installed. Both are surrounded the Tracker outer barrel (TOB, 6 layers) which is 220 cm long and provides z coverage up to 118 cm. In barrel and endcap regions, the pitch of strips changes from $80\mu\text{m}$ to $120\mu\text{m}$ and $100\mu\text{m}$ to $141\mu\text{m}$ respectively. Finally, the most outer part of CMS tracker system is tracker outer barrel (TOB) installed at radial distance from 55cm to 116cm containing $500\mu\text{m}$ thick 6 barrel layers with strip pitch of $183\mu\text{m}$ and $122\mu\text{m}$. Beyond $z=118 \text{ cm}$, there are tracker endcaps (TEC+, TEC-, 9 discs) which cover $124 \text{ cm} < |z| < 282 \text{ cm}$ and $22.5 \text{ cm} < |r| < 113.5 \text{ cm}$ providing coverage of $|\eta| < 2.5$.

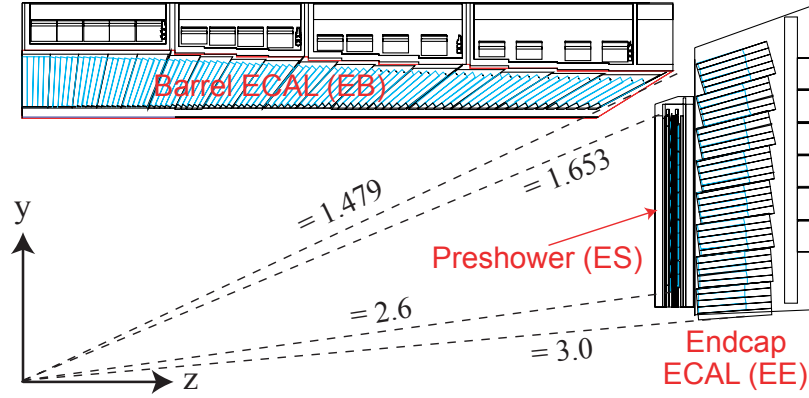


FIGURE 1.7: CMS ECAL pseudo-rapidity coverage.

1.1.4 Electromagnetic Calorimeter (ECAL)

ECAL is the cylindrical shaped homogeneous calorimeter built with 75848 crystals of lead tungstate (PbWO_4) placed inside the CMS solenoid magnet. Its barrel region (EB) contains 61200 PbWO_4 crystals and provides pseudorapidity coverage of $|\eta| < 1.479$. Endcap region of the ECAL (EE) is built with 7324 crystals on either side in endcap regions and provides pseudorapidity coverage up to $|\eta| < 3$. The region of $1.4442 < |\eta| < 1.56$ is regarded as crack region between barrel and endcaps as can be seen in Figure 1.7 taken from [Reference34]. This region is excluded for physics analysis.

When high energy particle (e.g. electron or photon) coming out of the collision point enters ECAL, it produces bunch or cascade of particles (electron through bremsstrahlung and photon through pair production) with relatively lower energies. This is called EM showering that continues until electrons dominate in ionization and photon energy becomes less than pair-production threshold.

Photodetectors Avalanche photodiodes (APDs) in barrel region and the Vacuum phototriodes (VPTs) in endcap regions are installed which efficiently work under high magnetic field environment [cms:ECALTDR]. Response of ECAL comes with emission of light that is directly proportional incident particle's energy (through ionization).

A preshower detector is installed in endcap regions to distinguish between single photon and closeby diphoton candidate e.g neutral pion decay ($\pi^0 \rightarrow \gamma\gamma$). Providing pseudorapidity coverage of $1.653 < |\eta| < 2.6$, it helps to efficiently determine the position of an electron and a photon.

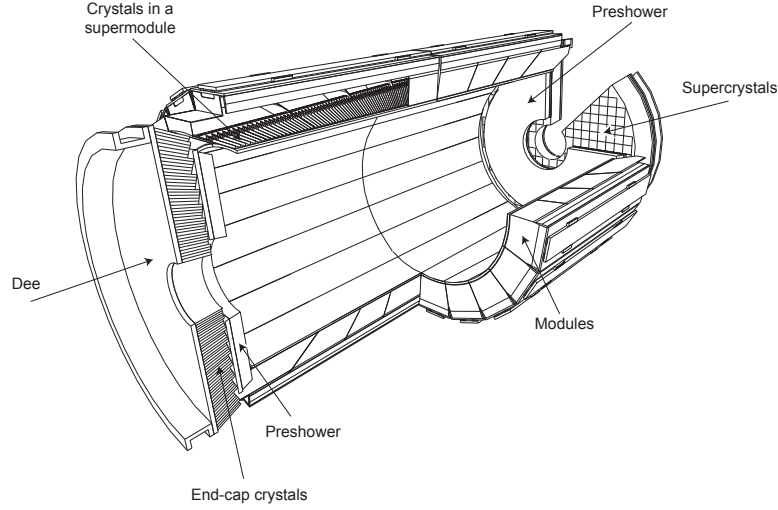


FIGURE 1.8: CMS ECAL layout

Lead tungstate is benefitted as it helps to make ECAL compact with high granularity due to the fact that it is denser (8.28 g/cm^3) with less radiation length (X_0 of 0.89 cm). Also, lead tungstate has small Molière radius, which defines the area of deposited energy, helping to minimize the effect of pileup while measuring the energy. Other important factor is that crystals of lead tungstate have decay time of scintillation about 25 ns which is also the time of LHC bunch crossing.

ECAL operating temperature is around 18°C and light emitted from the scintillation process is temperature dependent. So, in order to maintain the temperature at nominal value, cooling system is installed which is based on water circulation through Aluminium pipes and is capable enough to maintain the temperature of detector upto $18 \pm 0.05^\circ \text{C}$. Layout of the ECAL of CMS detector is shown in the Figure. 1.8.

Energy resolution of ECAL, which is the ability of detecting system to accurately measure energy of incident radiation, can be written as

$$\left(\frac{\sigma}{E}\right)^2 = \left(\frac{A}{\sqrt{E}}\right)^2 + \left(\frac{\sigma_n}{E}\right)^2 + c^2 \quad (1.2)$$

where A is stochastic term (defines the intrinsic statistical fluctuations in shower, in energy deposits of preshower, and the photo(electron)-statistics contributions), σ_n is the electronics noise term and c is constant that covers the constant factors such as non-uniformity in detector for light collection and etc. Contribution of

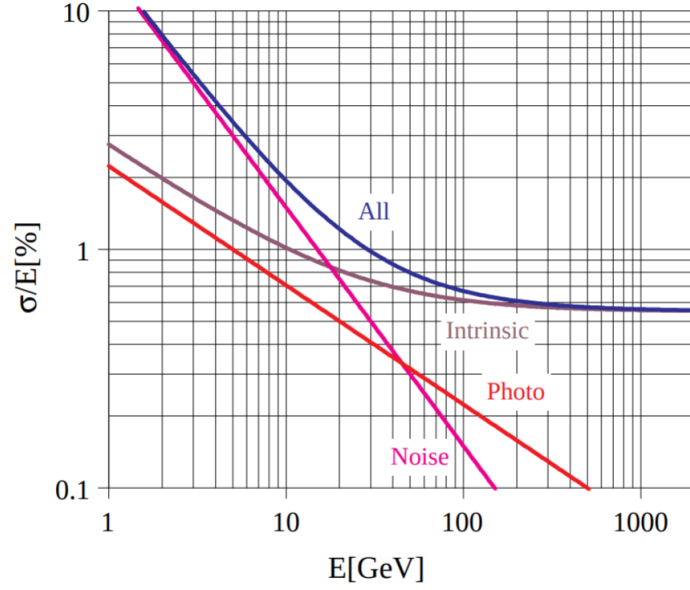


FIGURE 1.9: CMS ECAL energy resolution curve with individual term contributions.

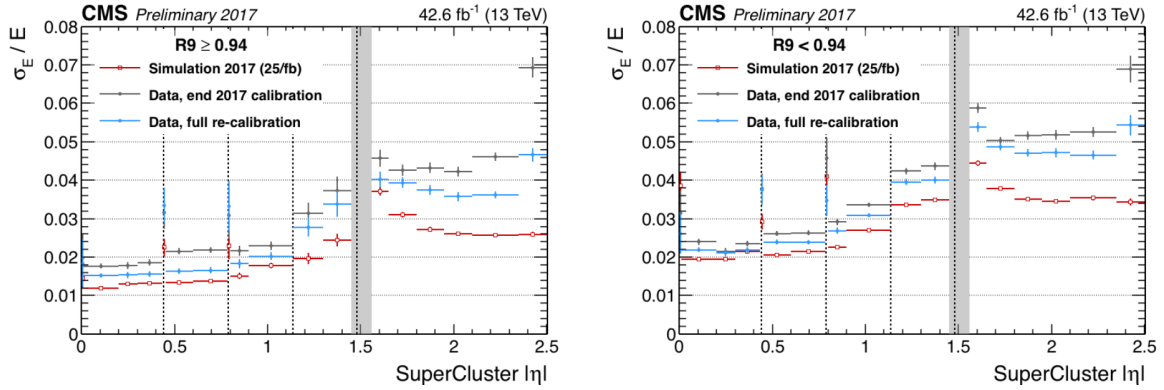


FIGURE 1.10: CMS ECAL energy resolution.

each term to the energy resolution is shown in the Figure. 1.9.

A recent study on relative energy resolution of ECAL using 2017 data and MC has been performed using Z events decaying to a pair of electrons. The resolution plots for bremsstrahlung electrons ($R_9 > 0.94$ and $R_9 < 0.94$) is separately shown for data and MC in the Figure 1.10 [Zghiche_2019]. For $H \rightarrow 4\ell$ analyses, excellent energy resolution for leptons is important for precise measurements for which ECAL of CMS plays an important role as it did for Higgs discovery.

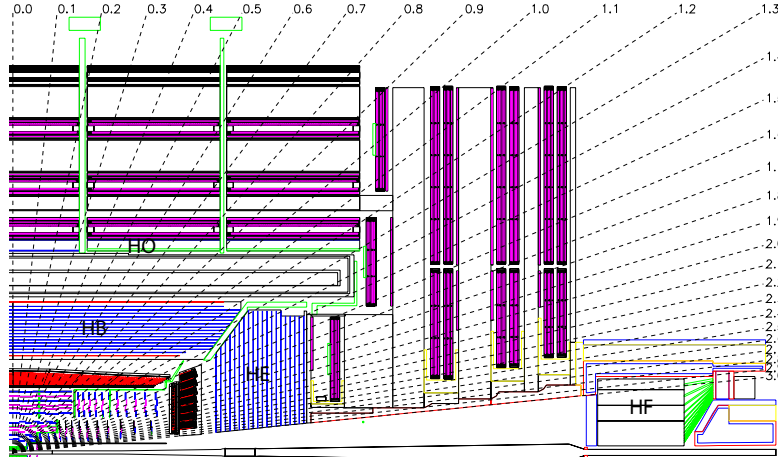


FIGURE 1.11: CMS HCAL layout.

1.1.5 Hadron Calorimeter (HCAL)

It is the part of CMS detecting system, which helps for the detection and as well as energy measurements of strongly interacting particles such as quarks (or the ones in composition of quarks), gluons hadronic jets (like neutrons, pions and etc.). Being hermetic, HCAL also helps to measure missing transverse energy (MET) which can be associated to invisible particles like neutrinos which is important studies like top quark production. HCAL assembles with CMS system between the solenoid magnet and ECAL as can be seen in the Fig. 1.1 at radial distance from 1.77 m to 2.95 m. Benefitting from pretty short interaction length, at most brass is chosen as absorber throughout the HCAL. Brass is a non-magnetic low cost material. HCAL provides pseudorapidity coverage upto $|\eta| < 3.0$ beyond which, 11.2 m away from the interaction point, forward Hadron Calorimeter (HF) is employed on both sides of the detector providing additional coverage up to $|\eta| < 5.2$. HF helps to detect particles from proton-proton collisions emerging at very small angles with respect to the beam axis.

Schematic diagram of HCAL is shown in the Figure 1.11 [Reference34].

HCAL is splitted into barrel and endcap regions, HB and HE which give pseudorapidity coverage of $|\eta| < 1.33$ and HE of $1.3 < |\eta| < 3$ respectively.

HCAL is also splitted into barrel and encap regions, HB and HE respectively. HB consists of 32 towers provides coverage of pseudorapidity of $|\eta| < 1.4$ and HE of $1.3 < |\eta| < 3$. This results in 2304 towers in a segmentation of $\Delta\eta \times \Delta\phi = 0.87 \times 0.87$.

HB serves as a single longitudinal sampling readout. HB is split into two halves (HB^+ and HB^-) which consist of 36 similar wedges each weighing 26 tonnes made up of brass plates. It may happen that tail of showers of hadrons which leak from rear of calorimeters may penetrate into ECAL and/or magnet. Therefore, additional calorimeter called Hadron Outer (HO) is installed outside the solenoid that used the coil of the magnet as absorber. With this effective thickness of HCAL becomes above 10 interaction lengths. HO has scintillator of 10 mm thickness and helps to improve resolution of HCAL for MET.

Pseudorapidity coverage of $3 < |\eta| < 3$ is provided by sub part of HCAL which is called Hadron Forward (HF). It suffers huge flux of the particles due to which sampling material is required to be hard against the radiations. For this purpose, steel/quartz fibre was used. HF is located 11.2 away from interaction point with 1.65 m absorber depth. With the incident particle flux, Cerenkov radiations are emitted through quartz fibers which is then channeled to photomultipliers (PMs).

HCAL is constructed with alternate layers of plastic and brass (the absorber and scintillator respectively) which helps to measure the arrival time, position and energy of a particle. Plates of brass are 79 mm thick each with 99 mm gaps for the scintillators installation. When a particle passes through scintillator, optical fibers collect the emitted lights and pass it to readout box. Total energy is calculated by summation of the light over many layers in depths called tower. The scintillator in HCAL are arranged in the form of 70,000 tiny tiles.

The energy resolution of the HCAL (i.e. HB, HO and HE) is given as

$$\left(\frac{\sigma}{E}\right)^2 = \left(\frac{90\%}{\sqrt{E}}\right)^2 + (a)^2 \quad (1.3)$$

and for that of HF it is given as

$$\left(\frac{\sigma}{E}\right)^2 = \left(\frac{170\%}{\sqrt{E}}\right)^2 + (b)^2. \quad (1.4)$$

Where a and b are constants having values 0.045 and 0.09 respectively.

There were subsequent upgrades in HCAL during Run 2 era. In 2017, PMTs were replaced with new multi-anode PMTs and additionally electronics replaced for better differentiation of real hit from the muons which hit the PMT window. In 2017, HE of HCAL was upgraded and photo detectors were replaced with latest silicon photo multipliers (SiPMs) for noise elimination and better photo detection

efficiency [pixel_hcal_upgrade].

1.1.6 CMS Muon System

Identification of muons originating from the collision point is a great challenge. Having long lifetime (i.e. 2.19×10^{-6} which is even higher in relativistic regime), they tend to cross the whole detector without decaying. Also, having higher mass (about 200 times the mass of electrons) it tends to lose very little energy via bremsstrahlung. So, they leave track however they deposit very little amount of energy in the calorimeters. Hence a dedicated muon system was proposed and installed as outermost sub-detector of CMS. It is benefitted from three different strategies for detection and measurements on muons.

- Drift Tubes (DT), only in barrel region
- Cathode Strip Chamber (CSC), only in endcap region
- Resistive plate chamber (RPC), in barrel and endcap regions both

Muon identification in the CMS detector is provided by its outer most layer of sub-detector called muon system [cms:muonsystem_TDR]. The detection of the muons is very important for many physics process. In particular, for SM Higgs boson decay to ZZ/ZZ^* which is labeled as a golden channel if each Z boson decays to a pair of opposite charged muons. Best mass resolution can be obtained with four muons in final state as muons are less effected by detector material as compared to electrons because they interact weakly with detector material. Along with this example, SUSY and other model for new physics searches also encourages the necessity for their reconstruction of muons with excellent resolution.

To have a robust and precise muon system was one of the core motive of CMS as indicated by detector name. Identification, triggering and momentum measurement of the muons are the tasks assigned to the muon system. The flux-return yoke and strong magnetic field are playing crucial role for excellent muons momentum resolution and triggering. The flux-return yoke by absorbing the hadrons helps for muons identification.

Muon system was essentially designed as cylindrical shape to provide full pseudorapidity coverage with detection area of 25000 m^2 , having barrel and end-caps regions. Muons detection is based on gasses detectors of three types. The

strength of magnetic field is uniform in barrel region with less neutron-induced background. Drift tubes are installed providing pseudorapidity coverage $|\eta| < 1.2$, as illustrated in Figure 1.12. The barrel region is arranged to have four muons station, measuring $r - \phi$ bending plane and distance along beam side. The arrangement of the drift cells are offset by the half of the width of cell providing dead spot in efficiency, determination of standalone bunch crossing and excellent time resolution by utilizing mean timer circuits. The direction and number of chambers are optimized to get maximum efficiency of combination of the hits from different stations to get single track which is helpful to reject the background. In endcap region, where non-uniform strength of magnetic field and more background is expected cathode strip chambers (CSCs) are installed that providing coverage $0.9 < |\eta| < 2.4$. Like DT, there 4 CSCs stations on each endcap. Each station is composed of chambers, which are capable to work independently, aligned perpendicularly to z-axis and spread between the flux-return plates. In order to overcome uncertainties of background rates and assignment of correct bunch crossing number, resistive plate chambers (RPCs) are planted both in the barrel and the endcap detector regions. There is a dedicated trigger system for each type of muons gasses detector including RPCs. Good time resolution, fast response and coarser spatial resolution makes RPCs distinct from CSCs and DT. The RPCs also plays vital role for the combination of multiple hits in a chamber to for tracks. The RPCs are installed in between DT and CSCs in region the barrel and the endcap region respectively.

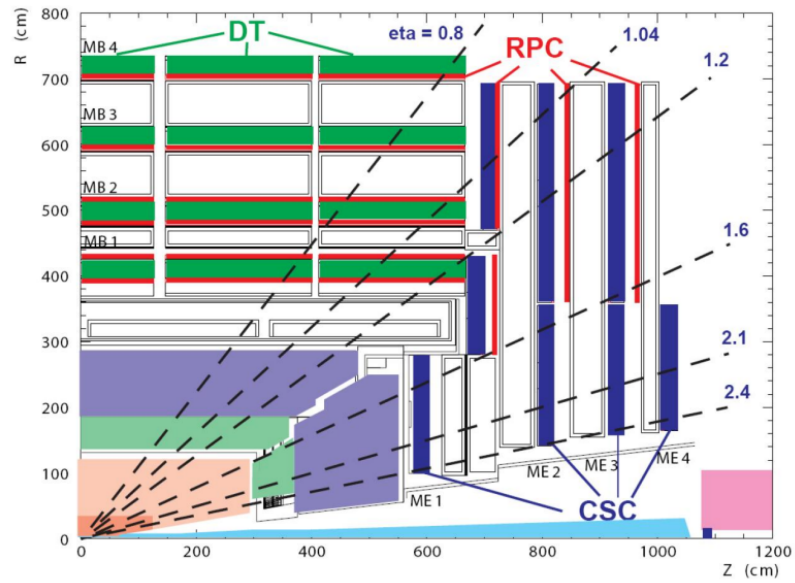


FIGURE 1.12: CMS muon system layout.

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